

RAMCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Academic Year: 2020- 2021 (Even Semester)

Date: 19.06.2021

Degree, Semester & Branch: II Semester B.E. Computer Science and Engineering

Course Code & Title: CS8251 Programming in C

Name of the Faculty member: Mr.R.Venkatesh, AP (SG)/CSE

Innovative practice: Active Learning- Crossword Puzzle

Topic: Structures and Dynamic Memory Allocation

Objectives of the activity:

O1	To familiarize students with the concepts of structures and dynamic memory allocation.
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Justification for choosing the topic:

In C programming, one of the most important concepts is structures and Dynamic Memory Allocation. Instead of a regular class teaching, interaction, and multiple-choice question quiz, a game-based event was created to help students remember and recollect the structure and dynamic memory allocation concepts.

Activity:

A crossword puzzle is an enjoyable word puzzle activity that helps students remember the concept and try to figure out the answer on their own. It is a game, so it is a more enjoyable activity than a regular multiple choice questions quiz.

For 20 minutes, the activity was carried out via the online platform tool Google meet. Canvas was used to distribute a crossword puzzle with 14 questions to the students. They must think about and provide answers to the given puzzle, and they need to submit the answers back to the canvas.

In the first 15 minutes of the revision class, the students answered the questions. They then submitted their answers to Canvas, where they were corrected. Figures 1 show the questions and picture of the crossword puzzle and 2 show examples of responses. The answers to the incorrect questions were discussed at the end of the class to ensure that they knew the correct answers to all of the questions.

ACROSS

1. Referential _____ Structures contains a reference to data of the same type
3. Which one is cannot be a structure member?
5. If a variable is a pointer to a structure, then which of the following operator is used to access data members of the structure through the pointer variable? (In letters)
7. VN); How will you free the allocated memory?
8. Memory is allocated for a structure when ____ declaration is done

DOWN

1. Presence of code like “st.b = 10” indicates _____
2. Which of the following operators can be applied on structure variables?
4. A structure is declared using the keyword struct followed by a structure _____
6. For int and float structure members, the values are initialized to _____

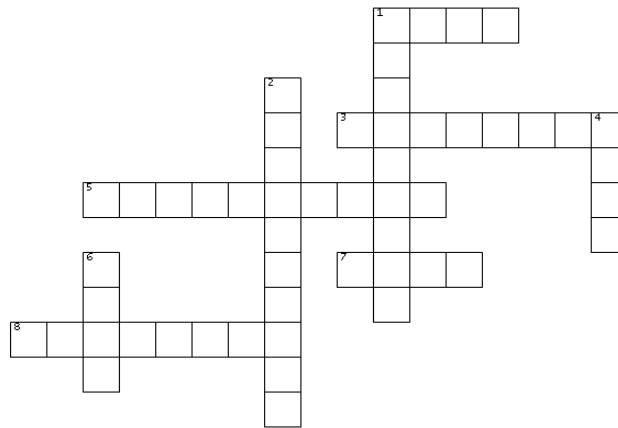


Figure1: Crossword Puzzle given to the students

ANSWER:

ACROSS

1. Self
3. Function
5. Pointing to
7. Free
8. Variable

DOWN

1. Structure
2. Assignment
4. Name
6. Zero

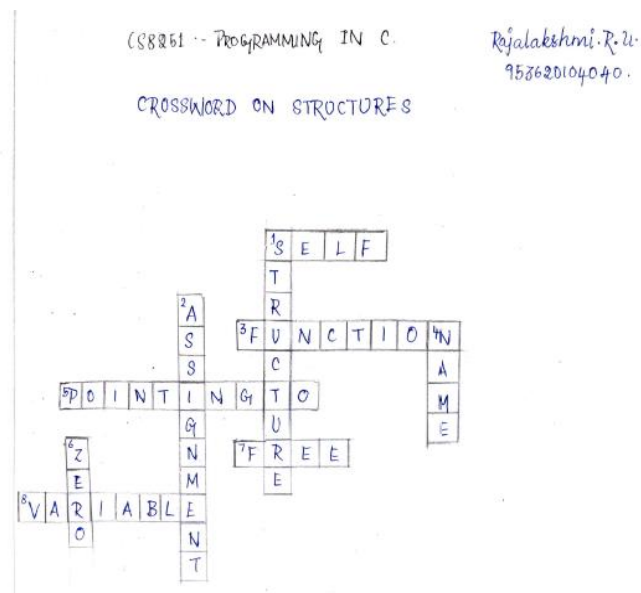


Fig 2: Sample Crossword Puzzle answer of Rajalakshmi R U

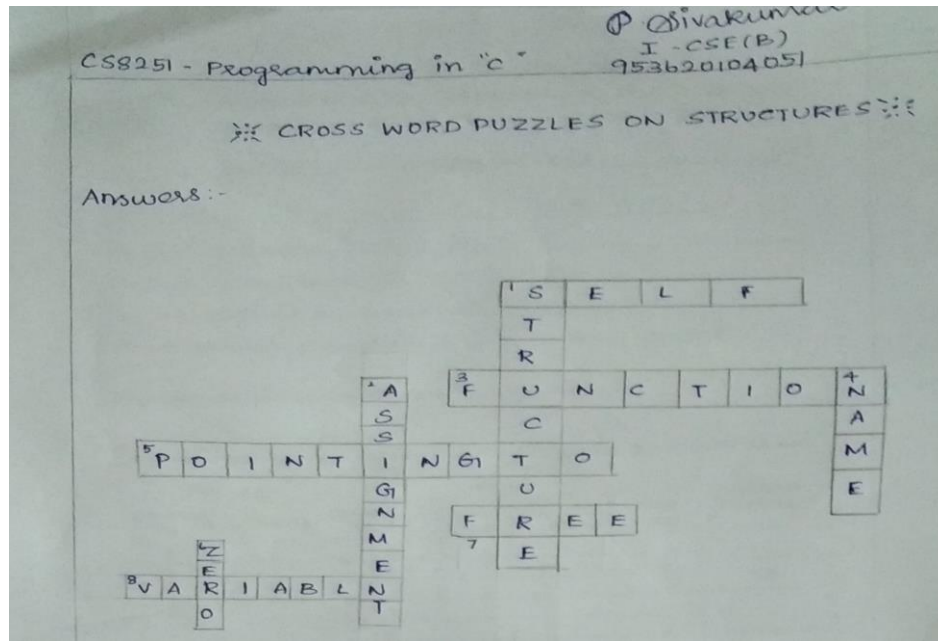
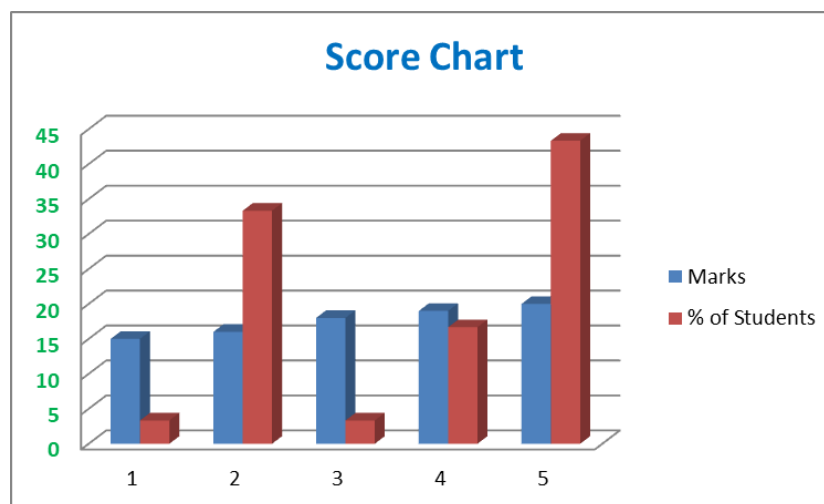


Fig 2: Sample Crossword Puzzle answer of Sivakumar P

Outcome:

Analysis was performed based on the answers provided by the students, and the results are presented in figure 3 as a chart. We can infer from this that the vast majority of students seemed to be familiar with the concepts of Structure and Dynamic memory allocation in C. This activity assisted students in thinking independently and recalling concepts.

Objectives/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	3	1	-	-	-	-	-	-	3	-	-	1



Reflective Report

Pre-implementation:

Identified Challenges:

- Few students do not recollect the concepts and they struggle to understand the puzzle.

Post implementation

- This game-like activity was enthusiastically participated in and enjoyed by all of the students.
- They can easily remember the basic of structures like nested structure, variables and members and dynamic memory allocation.

References:

1. <https://www.ritrjpm.ac.in/images/computer-science/Crosswords-MCAP-MGM.pdf>
2. <http://www.classtools.net/crossword/>
3. <https://www.teachingenglish.org.uk/article/using-flash-cards-young-learners>
4. <https://voxy.com/blog/2011/05/are-flashcards-an-effective-learning-tool-infographic/>
5. <https://elearningindustry.com/the-5-best-free-flashcard-creation-tools-for-teachers>
6. <https://www.education.com/worksheet-generator/reading/crossword-puzzle/>

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